

# Software Test Plan (STP)

**Project: Online Bookstore**

**Version: 1.0**

Authors: Group 2

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Status: Draft for Mini-Project Deliverable 1

A multi-seller e-commerce platform for buying and selling new and second-hand books.

*PES University, Bengaluru | UE24CS341A - Software Engineering | Semester V*

## Revision History

| Version | Date        | Author  | Change Summary                                                                                                              |
|---------|-------------|---------|-----------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 02-Sep-2026 | Group 2 | Initial test-plan structure adapted from the assigned STP template.                                                         |
| 0.5     | 05-Oct-2026 | Group 2 | Test scope, strategy, security validation, environment, traceability and test cases aligned with Online Bookstore SRS v1.0. |
| 1.0     | 05-Oct-2026 | Group 2 | Baseline STP prepared for Mini-Project Deliverables Part-1.                                                                 |

## 1. Introduction

**Purpose:** This Software Test Plan defines the testing objectives, scope, strategy, resources, schedule, responsibilities, risks and traceability for the Online Bookstore version 1.0. It is derived from the approved SRS and is intended to guide functional, non-functional, security and acceptance testing.

**Scope:** Testing covers account and access management; catalogue, search and reviews; cart, wishlist, checkout, payment and orders; seller operations; administrator moderation and reporting; and the performance, security, usability, maintainability and portability requirements stated in the SRS.

**Out of scope:** real-money settlement, physical logistics integration beyond recording courier tracking information, mobile application development, multi-language content, seller-to-seller bulk trade and machine-learning recommendation features, because these are explicitly outside SRS version 1.0 scope.

**References:** Online Bookstore SRS v1.0; Online Bookstore SAD v1.0; SRS Requirement Traceability Matrix; Appendix B field layouts and reports; IEEE Std 830-1998; OWASP Top 10 (2021).

**Definitions:** SRS (Software Requirements Specification), STP (Software Test Plan), RTM (Requirement Traceability Matrix), FR (Functional Requirement), NFR (Non-functional Requirement), JWT (JSON Web Token), RBAC (Role-Based Access Control), OTP (One-Time Password).

## 2. Test Items

- Authentication and user/account module - registration, OTP verification, login/logout, password recovery, profile/address management and RBAC.
- Catalogue and discovery module - catalogue pagination, keyword search, filtering, sorting, book details, reviews and related-title display.
- Cart, wishlist, checkout and order module - cart persistence, quantity validation, checkout revalidation, cancellation and return lifecycle.
- Payment integration - payment intent, signed callback validation, retry handling and idempotent callback processing.
- Seller operations - listing creation/editing, price/MRP validation, inventory, low-stock alerts, dispatch and seller dashboard.
- Administrator operations - moderation, user/listing/review blocking, audit logging and platform analytics/export.
- Notification integration - OTP, order confirmation, invoice and low-stock notifications with retry handling.
- Database and persistence - MongoDB collections, indexes, order-price freezing, ownership and data-integrity rules.
- Redis session/cache behaviour and performance-related caching.
- Responsive React user interface and cross-browser behaviour.

## 3. Features to be Tested

The following features are mapped directly to the SRS requirement groups. The detailed test cases in Appendix A provide executable coverage.

| Requirement group | Features / test focus                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------|
| FR-01–FR-06       | Registration, OTP, authentication, password recovery, profile/address management and RBAC.             |
| FR-07–FR-12       | Catalogue paging, indexed search, filtering/sorting, book detail, verified reviews and related titles. |
| FR-13–FR-18       | Cart, wishlist, checkout validation, payment, invoice, order lifecycle, cancellation and returns.      |
| FR-19–FR-24       | Seller listings/inventory/fulfilment/dashboard and administrator moderation/analytics.                 |
| NFR-01            | Search, catalogue/detail and order/report performance under the reference load.                        |
| NFR-02            | Application availability and health-check monitoring.                                                  |
| NFR-03            | Authentication, TLS, payment isolation, input validation, authorization and audit security.            |
| NFR-04            | Usability, 360px responsive checkout, WCAG 2.1 AA contrast and keyboard accessibility.                 |
| NFR-05            | Atomic stock decrement and prevention of overselling.                                                  |
| NFR-06            | ESLint quality gate and minimum 70% service-module unit-test coverage.                                 |
| NFR-07            | Compatibility with latest two major versions of Chrome, Edge, Firefox and Safari.                      |
| NFR-08            | Append-only audit logging for administrative actions and payment events.                               |

## 4. Features Not to be Tested

- Real-money payment settlement; the SRS restricts payment to the gateway sandbox.
- Storage or processing of card data by the Online Bookstore; card data remains with the payment gateway.
- Physical courier/logistics execution; the system records courier and tracking information only.
- Mobile application implementation; mobile-browser responsiveness is tested, but a native mobile application is out of scope.
- Multi-language content.
- Seller-to-seller bulk trade.
- Machine-learning recommendation engine.
- Dedicated hardware interfaces, because the SRS states that the product uses commodity cloud containers and standard browsers.

## 5. Test Approach / Strategy

### 5.1 Test Levels

- Unit Testing: Test individual service modules, validation rules and utility functions in isolation. Service dependencies such as payment gateway and SMTP should be injectable/stubbable.
- Integration Testing: Verify interactions between React/API, Express services/MongoDB, Redis, payment gateway adapter, SMTP adapter and object storage interface.
- System Testing: Execute complete user journeys such as registration → verification → login, search → cart → checkout → payment, seller listing → fulfilment and administrator moderation.
- Acceptance Testing (UAT): Confirm that the implemented system satisfies the demonstrable SRS behaviour and course acceptance expectations.

### 5.2 Test Types

- Functional testing - verify FR-01 to FR-24.
- Regression testing - rerun impacted tests after defect fixes or changes.
- Performance testing - verify NFR-01 response targets and reference load behaviour.
- Security testing - verify authentication, authorization, TLS, validation, payment isolation, rate limiting and audit controls.
- Usability/accessibility testing - verify NFR-04, responsive behaviour, keyboard access and readable error feedback.
- Compatibility testing - verify NFR-07 across the specified browsers.
- Data-integrity testing - verify NFR-05, including concurrent purchase of the last available copy.

### 5.3 Test Design Techniques

- Equivalence partitioning for valid/invalid registration, quantity, price, rating and filter inputs.
- Boundary-value analysis for password length, OTP validity, quantity limits, rating 1–5, review length and return-window rules.
- Decision-table testing for RBAC, payment outcomes, order cancellation/return and listing status.
- State-transition testing for order states PLACED → CONFIRMED → PACKED → SHIPPED → DELIVERED and CANCELLED/RETURNED paths.
- Negative testing for unauthorized routes, invalid payment callbacks, invalid ownership and malformed input.

### 5.4 Entry Criteria

- A testable build is available and deployable.
- Required test data and seeded catalogue are available.
- MongoDB/Redis test environment and external service sandboxes are reachable.
- Critical interfaces have stable contracts sufficient for integration testing.
- Test cases and expected results have been reviewed.

### 5.5 Exit Criteria

- 100% of planned test cases executed.
- All critical and blocker defects are closed or formally accepted by the project team/faculty.
- All high-priority SRS requirements have at least one passing test.
- Security validation activities in Section 5.1 are completed.
- Performance and data-integrity acceptance targets are demonstrated or documented with results.
- Requirement traceability coverage is complete for the selected test scope.

### 5.6 Test Case Priority

| Priority | Meaning                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------------------|
| High     | Failure prevents a core buyer/seller/admin journey or violates a critical security/data-integrity requirement. |
| Medium   | Failure affects a supporting function but core transaction flow can continue.                                  |
| Low      | Cosmetic, convenience or lower-impact behaviour.                                                               |

## 5.1 Security Validation

Security validation is based on the SRS security requirements and the Online Bookstore security architecture. Testing must demonstrate that security controls are enforced server-side rather than relying only on client-side presentation.

- Authentication validation: verify bcrypt password handling, verified-account requirement, JWT access-token expiry and refresh-token invalidation on logout/password change.

- OTP validation: verify six-digit OTP, ten-minute validity, invalid-attempt handling and request-rate cap of three per hour per account.
- Authorization validation: attempt buyer access to seller/admin endpoints and verify HTTP 403; verify object-level ownership for orders, addresses, carts, listings and settlements.
- TLS validation: verify client-server communication uses TLS 1.2 or higher and plain HTTP is redirected to HTTPS.
- Payment security: verify that card data is never stored/logged by the application; verify payment webhook signature validation and idempotent duplicate-callback handling.
- Input security: test malformed and injection/XSS-oriented inputs against server-side validation and sanitisation.
- Rate limiting: verify repeated failed logins are rate-limited and OTP requests are capped.
- Audit validation: verify administrative actions and payment events record actor, action, target and timestamp in an append-only audit log.
- Session/security review: verify access-token expiry, refresh-token rotation and blocked-user rejection at authentication.

## 6. Test Environment

| Area               | Planned environment                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client             | React single-page application accessed using the latest two major versions of Chrome, Edge, Firefox and Safari.                                                                               |
| Responsive devices | Desktop browser plus mobile-browser viewport down to 360 px as specified by the SRS.                                                                                                          |
| Server             | Node.js 20 LTS on a Linux container / managed cloud host.                                                                                                                                     |
| Database           | MongoDB 7 on managed MongoDB Atlas.                                                                                                                                                           |
| Cache / sessions   | Redis 7.                                                                                                                                                                                      |
| Payment            | Payment gateway sandbox REST API.                                                                                                                                                             |
| Notification       | SMTP service over TLS.                                                                                                                                                                        |
| Images             | Cloud object storage interface.                                                                                                                                                               |
| API                | HTTPS REST API with JSON and bearer JWT.                                                                                                                                                      |
| Test data          | Seeded catalogue of 10,000 listings; user roles Buyer/Seller/Administrator; controlled order, payment and inventory records.                                                                  |
| Performance load   | Reference load of up to 200 concurrent users as stated in SRS.                                                                                                                                |
| Tools              | GitHub/CI and OpenAPI/Swagger are project-supported tools. Postman and a performance/load tool may be used for execution where available; exact tool selection is an implementation decision. |

## 7. Test Schedule

| Milestone                             | Planned date | Output                                              |
|---------------------------------------|--------------|-----------------------------------------------------|
| Test-plan preparation                 | 05-Oct-2026  | STP baseline aligned to SRS/SAD.                    |
| Test-case design and review           | 05-Oct-2026  | Functional, non-functional and security test cases. |
| Environment/data setup                | 05-Oct-2026  | Test environment and seed data ready.               |
| Functional/system execution           | 06-Oct-2026  | Executed test cases and defect records.             |
| Security validation                   | 06-Oct-2026  | Security validation results.                        |
| Performance/data-integrity validation | 06-Oct-2026  | NFR-01 and NFR-05 evidence.                         |
| Final regression/UAT                  | 07-Oct-2026  | Regression results and acceptance evidence.         |
| Test summary                          | 07-Oct-2026  | Final Test Summary Report.                          |

## 8. Test Deliverables

- Software Test Plan (this document).

- Functional, non-functional and security test cases.
- Test data / seeded data required for execution.
- Test execution results and logs.
- Defect reports with severity, evidence and status.
- Requirement Traceability Matrix linking SRS requirements to tests.
- Performance/security validation evidence.
- Final Test Summary Report.

## 9. Roles and Responsibilities

| Role                             | Name                  | Responsibility                                                                             |
|----------------------------------|-----------------------|--------------------------------------------------------------------------------------------|
| QA / Test Lead                   | Group 2               | Maintain STP, coordinate execution, review coverage and consolidate results.               |
| Test Engineer                    | Group 2               | Design and execute test cases, record evidence and log defects.                            |
| Developer / Module Owner         | Group 2               | Support test environment, analyze defects, implement fixes and provide unit-test evidence. |
| Product Owner / Faculty Reviewer | Course team / faculty | Review acceptance results and provide final readiness/sign-off.                            |

SRS module ownership used for test coordination: Chenchu Sreekrishna Koushik - Identity/Accounts; Deeptha S - Catalogue/Search/Reviews; G H Pramod - Cart/Checkout/Payment/Orders; Hardhick M Gowda - Seller/Admin/Reporting.

## 10. Risks and Mitigation

| Risk                                                       | Impact                                                  | Mitigation                                                                                          |
|------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Unstable build delivered late                              | Reduces execution time and coverage.                    | Request early smoke builds and prioritize high-risk/high-priority test cases.                       |
| Payment sandbox unavailable                                | Checkout success/failure flow cannot be fully executed. | Use controlled test doubles/stubs for service-level tests and execute sandbox tests when available. |
| SMTP unavailable                                           | OTP and notification tests are blocked.                 | Stub notification adapter for service tests and retry integration tests when service is restored.   |
| Performance environment differs from reference environment | Measured response times may not be comparable.          | Record environment configuration with every performance result and use the SRS reference load.      |
| Insufficient test data                                     | Boundary and ownership tests cannot be executed.        | Prepare role-specific users, listings, stock states, orders and payment records before execution.   |
| Defect fixes cause regression                              | Previously working functionality may break.             | Run targeted regression plus the complete high-priority smoke suite after significant fixes.        |
| Third-party free-tier/resource limits                      | Test environment may become unavailable or slow.        | Monitor usage and keep external services within project free-tier constraints.                      |

## 11. Assumptions & Dependencies

- The Online Bookstore SRS v1.0 remains the baseline for test scope.
- The SAD architecture is implemented sufficiently for integration and system testing.
- MongoDB Atlas and Redis are reachable during test execution.
- The payment gateway sandbox is available and its callback behaviour is sufficiently representative for validation.
- SMTP service is available for integration testing or is replaced by a controlled stub for service-level tests.

- A seeded catalogue of 10,000 listings and a reference load of up to 200 concurrent users are available for performance validation.
- Sellers provide honest book-condition information; the system records the declared condition but cannot physically verify it.
- Courier dispatch itself is outside the system; only courier and tracking information are tested.

## 12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than four hours or becomes unusable.
- The build is sufficiently unstable that more than 30% of planned tests are blocked.
- A critical security or data-integrity defect makes further execution unsafe or invalid.
- A required external sandbox dependency is unavailable and no approved test stub is available.

Resume testing if:

- Blocking/critical defects are resolved or an approved workaround is available.
- The test environment and required services are stabilized.
- A new build passes smoke tests.
- Required test data and interfaces are restored.

## 13. Test Case Management & Traceability

The Requirement Traceability Matrix (RTM) maps SRS requirements to architecture/design references and test cases. The test suite below intentionally covers functional, non-functional and security requirements. The complete project RTM remains the authoritative source for requirement-to-code-to-test mapping.

| Requirement     | Representative test cases |
|-----------------|---------------------------|
| FR-01–FR-03     | TC-01, TC-02, TC-03       |
| FR-06           | TC-04                     |
| FR-07–FR-10     | TC-05, TC-06              |
| FR-11           | TC-07                     |
| FR-13–FR-16     | TC-08, TC-09              |
| FR-18           | TC-10                     |
| FR-19–FR-21     | TC-11                     |
| FR-23–FR-24     | TC-12                     |
| NFR-01          | TC-13                     |
| NFR-03          | TC-03, TC-04, TC-14       |
| NFR-04 / NFR-07 | TC-15                     |
| NFR-05          | TC-16                     |
| NFR-08          | TC-12, TC-14              |

## 14. Test Metrics & Reporting

### 14.1 Metrics

- % test cases executed =  $\text{executed test cases} / \text{planned test cases} \times 100$ .
- % test cases passed and failed.
- Requirement coverage =  $\text{requirements with at least one executed test} / \text{requirements in scope} \times 100$ .
- Defect density and defect severity distribution.
- Defect aging - time defects remain open by severity.
- Performance measures including 95th-percentile search response time and order confirmation time.
- Security validation pass/fail status.

- Regression pass rate after defect fixes.

## 14.2 Reports

- Daily execution status during active testing.
- Defect report / defect dashboard.
- Security validation report.
- Performance test result summary.
- Final Test Summary Report containing execution, pass/fail, defects, coverage and acceptance status.

## 15. Approvals

| Role                    | Name    | Signature / Date |
|-------------------------|---------|------------------|
| QA / Test Lead          | Group 2 | _____            |
| Development Lead        | Group 2 | _____            |
| Product Owner / Faculty | _____   | _____            |

## Appendix A: System Test Cases

The assignment requires sufficient test cases covering functional and non-functional requirements. The following 16 cases provide broader coverage than the minimum 7–10 requested and are directly aligned with the Online Bookstore SRS.

| TC ID | Test Case                          | SRS Ref.             | Priority | Precondition / Action                                                                   | Expected Result                                                                                 | Status  |
|-------|------------------------------------|----------------------|----------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------|
| TC-01 | Registration with valid data       | FR-01                | High     | Enter valid buyer registration data.                                                    | Account created unverified; six-digit OTP sent.                                                 | Pending |
| TC-02 | OTP verification                   | FR-02                | High     | Enter correct OTP within ten minutes.                                                   | Account becomes verified and can sign in.                                                       | Pending |
| TC-03 | Login and token handling           | FR-03, NFR-03        | High     | Login with verified valid credentials; inspect expiry/logout.                           | Access token expires in 15 min; refresh token follows policy; logout invalidates refresh token. | Pending |
| TC-04 | RBAC unauthorized access           | FR-06, NFR-03        | High     | Call seller/admin endpoint using buyer credentials.                                     | Request rejected with HTTP 403 and security/audit handling occurs.                              | Pending |
| TC-05 | Catalogue pagination               | FR-07                | Medium   | Open catalogue with seeded books.                                                       | Twenty titles are returned per page with required summary fields.                               | Pending |
| TC-06 | Search and filtering               | FR-08, FR-09         | High     | Search by title/author/ISBN/publisher and apply filters.                                | Correct ranked/filter results returned within performance target.                               | Pending |
| TC-07 | Verified-purchase review           | FR-11                | Medium   | Attempt review before delivery, then after DELIVERED.                                   | Pre-delivery review rejected; delivered purchase can submit one 1–5 rating.                     | Pending |
| TC-08 | Cart quantity and persistence      | FR-13                | High     | Add listing, change quantity, logout/login, return to cart.                             | Quantity obeys stock and 1–10 limits; cart persists across sessions.                            | Pending |
| TC-09 | Checkout and payment success       | FR-15, FR-16, FR-17  | High     | Checkout valid cart and complete sandbox payment.                                       | Order confirmed only after verified success callback; stock decremented; invoice generated.     | Pending |
| TC-10 | Order cancellation/return          | FR-18                | High     | Cancel before dispatch; attempt return after delivery.                                  | Cancellation accepted before dispatch; return path available within seven days of delivery.     | Pending |
| TC-11 | Seller listing/inventory           | FR-19, FR-20, FR-21  | High     | Create listing, set stock/price, dispatch an order.                                     | Price cannot exceed MRP; low-stock/out-of-stock rules and shipment record work.                 | Pending |
| TC-12 | Administrator moderation/reporting | FR-23, FR-24, NFR-08 | High     | Block listing/user and generate monthly report.                                         | Object is removed from buyer view, reason/action is audited, report generated/exported.         | Pending |
| TC-13 | Search performance                 | NFR-01               | High     | Run search under reference load of 10,000 listings/200 concurrent users.                | 95th-percentile catalogue search is within 2 seconds and indexed query plan is used.            | Pending |
| TC-14 | Security validation suite          | NFR-03, NFR-08       | High     | Test TLS, input validation, rate limiting, payment callback signature and audit events. | Security controls pass; invalid/unauthorized requests are                                       | Pending |

|       |                                                |                |        |                                                                    |                                                                                                          |         |
|-------|------------------------------------------------|----------------|--------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------|
|       |                                                |                |        |                                                                    | rejected; events are auditable.                                                                          |         |
| TC-15 | Responsive/accessibility/browser compatibility | NFR-04, NFR-07 | Medium | Run buyer flow at 360px and supported browsers; keyboard navigate. | Checkout usable within four screens; controls keyboard reachable; supported browsers function correctly. | Pending |
| TC-16 | Concurrent last-stock purchase                 | NFR-05         | High   | Two concurrent orders attempt to purchase the last copy.           | Exactly one succeeds in decrementing the final stock; listing is not oversold.                           | Pending |

## Appendix B: Test Case Execution Record

| TC ID | Actual Result                 | Pass/Fail | Defect ID | Executed By | Date |
|-------|-------------------------------|-----------|-----------|-------------|------|
| TC-01 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-02 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-03 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-04 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-05 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-06 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-07 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-08 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-09 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-10 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-11 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-12 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-13 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-14 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-15 | To be filled during execution | Not Run   | -         | -           | -    |
| TC-16 | To be filled during execution | Not Run   | -         | -           | -    |